Fourth National Level

RESEARCH CONCLAVE 2025

Theme:

"Advancing Societally Relevant Applications of Knowledge through Scientific Research"

May 15 -16, 2025

For any queries, contact:

Organizing Secretary

Research Conclave 2025

PSG College of Technology,

Peelamedu, Coimbatore-641004

https://www.psgtech.edu/rc2025/

res.conclave@psgtech.ac.in

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Organized by

PSG College of Technology

Coimbatore 641004

About the College

PSG College of Technology (PSG CT), established in the year 1951 by PSG & Sons' Charities, is an ISO 9001 - 2015 certified autonomous college affiliated to Anna University, Chennai. The college offers 21 undergraduate programmes and 24 postgraduate programmes including Engineering & Technology, computer applications, management sciences, basic and advance sciences. PSG CT is equipped with several state-of-the-art Centers of Excellence that include TIFAC-Core in Product Design, Machine Tool Research Centre, Engineering Design Centre, Virtual Reality Centre, Tool and Die Centre, Centre for Nano-technology, Centre for Robotics, Centre of excellence in Artificial Intelligence and Software, Centre for Non-Linear Dynamics, Danfoss Centre of Excellence in Climate and Energy, Centre of Excellence in Welding Engineering and Technology. Under the banner of PSG Industrial Institute, there are in-campus manufacturing units of machine tools, pumps, motors and off-campus foundry units. Furthermore, PSG CT has established a very good network with industry, research institutes, alumni and entrepreneurs. PSG CT was ranked second under Colleges/Institutes (Govt. & Govt. Aided) (Technical) by ATAL Ranking of Institutions on Innovation Achievements (ARIIA), Ministry of Education, Government of India in the year 2021 and the best industry linked institution by AICTE-CII in the year 2012. During the occasion of India assuming the G20 Presidency on December 1, 2022, PSG CT was identified as one among the 75 educational institutions across India, with special responsibilities towards organizing special lectures, student exchange programmes, academic and cultural activities to spread awareness among the youth and the academic community on the G20 agenda.

About Research Conclave 2025

Advancements in science and technology are essential for driving research development toward higher goals. Achieving these goals requires a multidisciplinary approach—a fusion of knowledge and insights from various disciplines and perspectives, all working toward the common objective of achieving a balance between scientific discoveries and technological applications. The program showcases cutting-edge scientific innovations across multiple fields, including academic research and industrial applications, with a particular focus on material development in a global context. This Research Conclave has been conceptualised to bring together leading researchers, renowned experts from various science and engineering fields, industry pioneers, and thought leaders. The primary aim of the conclave is to foster interaction and collaboration among researchers and professionals. The unique properties of materials are explored through expert discussions and well-established methodologies. We have made concerted efforts to unite diverse research groups from various engineering disciplines to develop novel material perspectives, design solutions for innovative applications, advance product technologies and device manufacturing. The impact of this field continues to influence many related disciplines.

This year's conclave will focus on critical areas such as Robotics and Automation, Additive Manufacturing Machine Learning, Logistics & Supply Chain Auotomation, Quantum & Neuromorphic Computing, 5G/6G and Mobile Networks, Analog & Digital Electronics, Bio-materials and many more. Given the scope of these topics, the conclave will undoubtedly provide a dynamic platform for interdisciplinary discussions amongst the experts and the delegates alike.

CALL FOR PAPERS

Participants from Academia, R&D organizations and Industries are invited to submit original and unpublished full length research papers on the following topics, but not limited to

Mechanical Sciences

- Machine Learning for Predictive Maintenance
- Life Cycle Assessment in Engineering Design
- Precision Engineering
- Additive Manufacturing
- Industry 5.0 and Resilient Manufacturing
- Engineering Materials and Processing

Automation and Digital Transformation

- Robotics and Automation
- Cyber Physical Systems
- Digital Transformation Strategies
- Logistics & Supply Chain Automation
- Intelligent Sustainable Manufacturing

Transformative Innovations in Computing

- Quantum and Neuromorphic Computing
- Artificial Intelligence, Machine Learning and Autonomous Systems
- Edge, Fog and Distributed Cloud Computing
- Blockchain, Cybersecurity and Secure Networking
- 5G/6G and Mobile Networks
- Augmented Reality, Virtual Reality and Human Augmentation

Advancements in Electronics and Intelligent Systems

- Analog and Digital Electronics
- Embedded Systems, IoT and Automotive Electronics
- Power Electronics, Drives and Energy-Efficient Systems
- Communication Systems and Signal/Image Processing
- VLSI, Sensors and Instrumentation

Advances in Biomaterials and Biotechnology

- Personalized Medicine and Biopharmaceutical Technology
- Biofuels, Bioprocessing and Bioremediation
- Bioinformatics and Computational Biology
- Biomedical Devices and Systems
- Biomaterials for Medical and Industrial Applications
- Biosensors

Reinforcement in Civil Engg. and Textiles Technology

- Sustainable Civil Engineering Structures
- Modern Construction and Building Materials
- Environmental Engineering
- Industrial / Healthcare Textiles
- Sustainable Fashion

Physical and Mathematical Sciences

- Material Synthesis and Characterization
- Computational Physics
- Organic/Inorganic Chemistry /Electrochemistry
- Mathematical Modelling and Optimisation
- Probability and Stochastic Processes
- Numerical and Statistical Analysis

Humanities and Entrepreneurship

- Entrepreneurial Business Model
- Literature and Human Values
- Economics and Management
- Linguistics, ELT and Contemporary English
- Market Competition and Disruptive Strategies

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Important Information

1. Abstract acceptance will be intimated through Email

2.Paper Submission & Author Guidelines visit https://www.psgtech.edu/rc2025/

3. The papers submitted will be checked for originality/plagiarism by the organizing institutions. (Accepted similarity: less than or equal to 15% for research papers)

4. Select papers presented in the conclave will be considered for possible publication in Scopus/ Web of Science (WoS) journals as per norms of the journals.

5. Industry participants are invited to submit short case studies on their R & D activities.

Payment of registration fee for Research Conclave 2025 can be made through online mode

Payment type	:	Internet Banking
Account Number	:	1481267367
Account Holder	:	PSG Centre for Non-formal and Continuing Education (PSG CNCE)
Bank Name	:	Central Bank of India
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IFSC Code	:	CBIN0280913

Dates to Remember

Full Paper Submission Deadline	:Apr 25, 2025
Notification of Acceptance	:Apr 28, 2025
Final submission	:May 02, 2025
Last Date for Registration fee	:May 02, 2025

Note:

A bonafide certificate is needed for external participants.

No registration fee for UG/PG/PhD Scholars from PSG College of Technology, Coimbatore.

Conclave Registration

Students	**Rs. 750/-**
Faculty members/Research Scholars	**Rs.1000/-**
Industry participants	**Rs.1500/-**

For Registration

RC 2022

RC 2024

Glimpses of Past CONCLAVE

RC 2023

Tentative Conference Schedule

Day 1 - May 15,2025		Day 2 - May 16,2025	
08:15 am - 09.00 am	Registration	09:00 am – 09:45 am	Keynote 4
09:00 am – 10:00 am	Inaugural function	09:45 am – 10:30 am	Keynote 5
10.00 am- 10.45 am	Keynote 1	11:00 am – 11:00 am	Refreshments
10:45 am – 11:00 am	Refreshments	11.00 am - 1.00 pm	Technical sessions in respective departments
11:00 am – 11:45 am	Keynote 2	1:00 pm – 02:00 pm	Lunch break
11:45 am – 12:30 pm	Keynote 3	03:00 pm- 04:00 pm	Valedictory function
12:30 pm – 01:45 pm	Lunch break	04:00 pm – 04:30 pm	Refreshments
01:45 pm – 03:00 pm	Technical sessions in respective departments		
03:00 pm – 03:30 pm	Refreshments		
03:30 pm – 04:30 pm	Technical sessions in respective departments		